

Message Exchange

The **Sigma1**, **Sigma2**, **Sigma3**, and **SigmaFinished** of a distinct session establishment are part of the same **message exchange**. The **Sigma1 with resumption**, **Sigma2_Resume** and **SigmaFinished** of a distinct session resumption are part of the same **message exchange**. The **Sigma1 with resumption**, **Sigma2**, **Sigma3** and **SigmaFinished** of a distinct session resumption that failed to perform the resumption are part of the same **message exchange**.

Each message SHALL use **PROTOCOL_ID_SECURE_CHANNEL** as **Protocol ID** and the corresponding **Protocol Opcode** as defined in Table 18, “Secure Channel Protocol Opcodes”.

The **Exchange Flags** of the **Protocol Header** are defined as follows per CASE message:

Message	I Flag
CASE Sigma1	1
CASE Sigma2	0
CASE Sigma3	1
CASE Sigma2_Resume	0

For the **SigmaFinished** message the value of the **I Flag** is set depending on whether the status message is sent by the Initiator or the Responder.

All CASE messages SHALL be sent reliably. This may be implicit (e.g. TCP) or explicit (e.g. **MRP** reliable messaging) in the underlying transport.

The other fields of the **Exchange format** are not specific to the CASE messages.

Generate and Send Sigma1

The initiator encodes and sends a **Sigma1** message, with a payload that follows this TLV schema:

```
sigma-1-struct => STRUCTURE [ tag-order ]
{
    initiatorRandom          [1] : OCTET STRING [ length 32 ],
    initiatorSessionId       [2] : UNSIGNED INTEGER [ range 16-bits ],
    destinationId            [3] : destination-identifier,
    initiatorEphPubKey       [4] : ec-pub-key,
    initiatorSessionParams   [5, optional] : session-parameter-struct,
    resumptionID             [6, optional] : OCTET STRING [ length 16 ],
    initiatorResumeMIC       [7, optional] : OCTET STRING [ length
CRYPTO_AEAD_MIC_LENGTH_BYTES ]
}
```

1. The initiator SHALL **generate a random number** **InitiatorRandom** = **Crypto_DRBG**(Len = 32 * 8).
2. The initiator SHALL generate a **session identifier** (**InitiatorSessionId**) for subsequent identification of this session. The **InitiatorSessionId** field SHALL NOT overlap with any other existing